Expression Purification and Crystallization of Magnetically Controllable Protein

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ABSTRACT

A novel magnetic field effect (MFE) on certain fluorescent proteins has recently been discovered where fluorescence intensity can be modulated in the presence of a magnetic field. MagLOV, a protein generated through directed evolution from a Light-oxygen-voltage (LOV) domain protein, is a new addition to the magnetic field sensitive fluorescent proteins (Hayward et.al https://zenodo.org/records/11406498). The LOV family of proteins exhibit conformational changes in response to light excitation and hence are often used to achieve optogenetic control of cellular functions (Lungu et.al Chem Biol 2012) (Pudasaini et.al Front. Mol. Biosci. 2015). In this project, we aim to perform X-ray crystallography on MagLOV in order to understand the structural basis of its MFE. To this end, we performed a large-scale expression and purification of MagLOV protein, verified MFE in purified proteins and performed extensive screening for possible crystal conditions. The potential structural changes upon light and magnetic field activation could guide us to further engineer MagLOV into a new type of "magnetogenetic" tool. This work has set the ground for structural and mechanistic studies for novel magnetic field sensitivity in LOV domain proteins.

I. INTRODUCTION

A. The LOV Domain Forms Fluorescent Adducts

The LOV domain is commonly used as a domain which Optogenetic tools are based off of. LOV domain proteins are notable for having a FMN chromophore which interacts with a Cysteine located in the GXNCRFLQ motif within the active site. This Cystine (C450) forms a

covalent bond with its sulfur and the C4 carbon of the FMN cofactor when excited with blue light. This formation is called the fluorescent adduct (Pudasaini et.al Front. Mol. Biosci. 2015).

B. AsLOV2 Structurally Responds to Light

The Optogenetic functionality of the LOV domain comes from the unfolding of a C terminus alpha helix known as the Jα helix when fluorescent activity occurs (Pudasaini et.al Front. Mol. Biosci. 2015) (Lungu et.al Chem Biol 2012) (Zayner et.al 2012). Within AsLOV2 this unfolding begins due to a change in the H bonding preference of key glutamine residue Q513 in response to the formation of the fluorescent adduct. This change was confirmed to occur through crystallographic (Nash et.al Biochemistry 2008) (Dietler et.al Nat Commun 2022) NMR and computational techniques (Iuliano et.al 2020).

C. Magnetic Field Effects In Nature

Magnetic Field Effects (MFE) have been observed in some capacity in nature for decades. The most obvious example of a natural response to a magnetic field is a bird's sense of magnetoreception. A model has been put forward suggesting that the underlying cause of this effect are the energetic interactions between Cryptochrome 4 Tryptophans and FAD, a flavin molecule, found within the eye of birds (Ritz et.al Biophys J 2000). In the 1970s and 1980s, studies were done to investigate the effect magnetic fields had on the chemistry of some photosynthetic reaction centers. These studies revealed a correlation between the activity of these reaction centers and the influence of a magnetic field (Boxer Biochimica et Biophysica Acta (BBA) - Reviews on Bioenergetics 1982).

D. MagLOV Exhibits Novel MFE

Where MagLOV differs from previous MFE expressing biomolecules is that it is a LOV domain protein which exhibits a unique effect where under the influence of a 10 mT magnetic field, MagLOV's fluorescent activity has been reported to lower by 75 %. This effect has been demonstrated in both live cells and in small batches of purified protein (Hayward et.al https://zenodo.org/records/11406498). MagLOV is a derivative of AsLOV2, with mutations C450P L496V Q513K D540M and G528K (Ingaromo and Coworkers https://zenodo.org/records/11406498). In this project, we aim to perform X-ray crystallography to elucidate the structure of MagLOV in its dark, light, MFE affected and MFE unaffected states.

II. METHODS

A. Optimization of MagLOV expression from pRSET Vector

pRSET-MagLOV plasmid was generously provided by Dr Maria Ingaromo at Calico Life Sciences. Plasmid was transformed into Thermo Scientific™ BL21 (DE3) E.coli and plated on 100 µg/mL ampicillin LB-Agar plates and incubated for 16 h at 37 °C. Up to 3 colonies from these plates were then added to primary LB Broth cultures containing 500 µM ampicillin and allowed to grow at 37 °C, 250 rpm for 16 hours. These then underwent 20X dilution into secondary cultures, where OD 0.6 would be reached between 6 and 8 hours in a shaker at 37°C 200 rpm. 400 µM of IPTG was added at OD 0.6 and cultures were induced in 16°C shakers at 200 rpm for 16 hours.

Similar results were achieved by adding primary culture or a transformed colony, prepared in the same way as in the first method, directly to large LB-Broth cultures and allowed to incubate at 16 °C for 63 hours. In order to check MFE of living cells, pRSET-MagLOV BL21

(DE3) cells were also plated on LB-Agar plates containing 100 µg/mL ampicillin and 10µM IPTG. A graphical representation of all three methods are shown in figure 1.

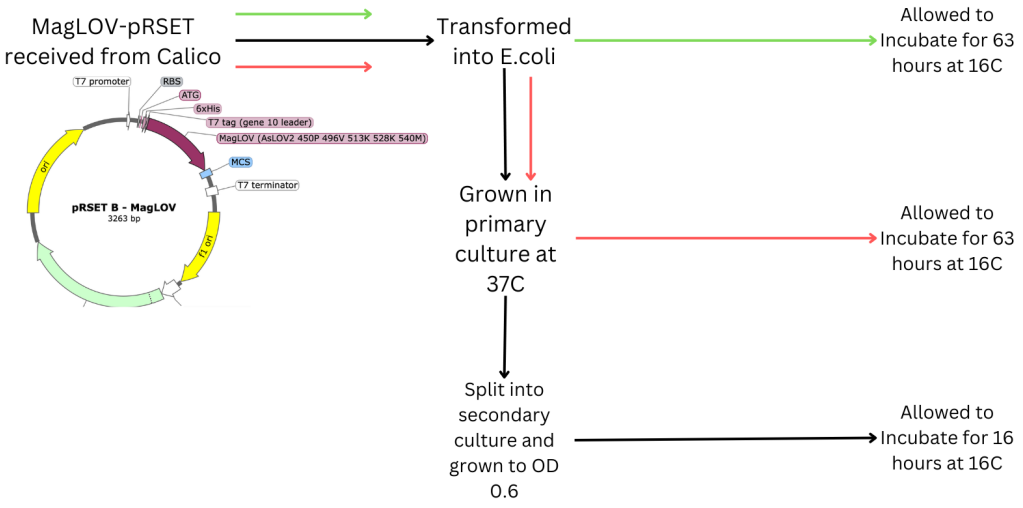


Figure 1: A general timeline illustrating how we express pRSET-MagLOV. The 3 primary methods we used are depicted via different colored arrows.

B. Expression of MagLOV in pET Vector

For the purpose of producing large volumes of protein, a MagLOV construct within a pET vector was used. We used a similar expression strategy as discussed above.

C. Purification of MagLOV

MagLOV is a HIS tagged protein and was separated from the bacterial lysate using Qiagen™ Ni-NTA Agarose Beads. Proteins from both pRSET and pET plasmids underwent this process. pET plasmid was then purified further through size exclusion chromatography into a storage buffer using the Bio Rad NGC Chromatography System.

D. HIS Tag Cleavage, Concentration and Desalting

A Thrombin cleavage site was also added to the insert sequence of the pET vector, and the HIS tag was cleaved using a Thrombin CleanCleave™ Kit from Millipore Sigma. Concentration was performed in Amicon® Ultra-15 Centrifugal Filter Units and Amicon® Ultra-4 Centrifugal Filter Units. Desalting was performed in Thermo Fisher Zeba™ Spin Desalting Columns.

E. MFE Measurements

MFE measurements were taken with the help of Dr. Shaun C. Burd, Physics Department, Stanford University. This custom built MFE measurement apparatus consists of a stage on which samples were loaded and irradiated with a blue light laser from above, while an electromagnet sits underneath the stage. This electromagnet is capable of generating field strengths of 30-70 mT, although real effective magnetic field strength is quite variable due to the interference of the loading stage. A camera fitted with a blue light filter is mounted above the sample stage and takes a video of the sample. A computer program then counts the amount of bright pixels and reports this out as a count graphed against time. The magnet was turned on and off for 20 second intervals to expose the protein to the magnetic field.

F. Protein Crystallization

Three attempts at protein crystallization have been performed so far. The first was based on the crystallization conditions used to crystalize wild type AsLOV2. The protein used in these trials was expressed from pET, purified with Ni-NTA, SEC, and desalted, and wells contained protein at 40 mg/ml and 20 mg/ml.

After this, a large crystal screen of 192 was attempted with protein which had undergone HIS tag cleavage in addition to Ni-NTA and SEC purification. There was another crystal screen of 288 conditions with desalted His Tagged protein,

III. RESULTS

A. Expression and Purification of MagLOV from pRSET and pET Vectors

Before an X-ray crystallization can be carried out, we must first generate crystals. Before crystals are formed we must have a high yield of pure protein, and confirmation that we have the correct protein by reproducing Calico's work. To this end, we expressed MagLOV with two vectors: pRSET and a pET Vector, and purified it through a two step Ni-NTA and SEC process.

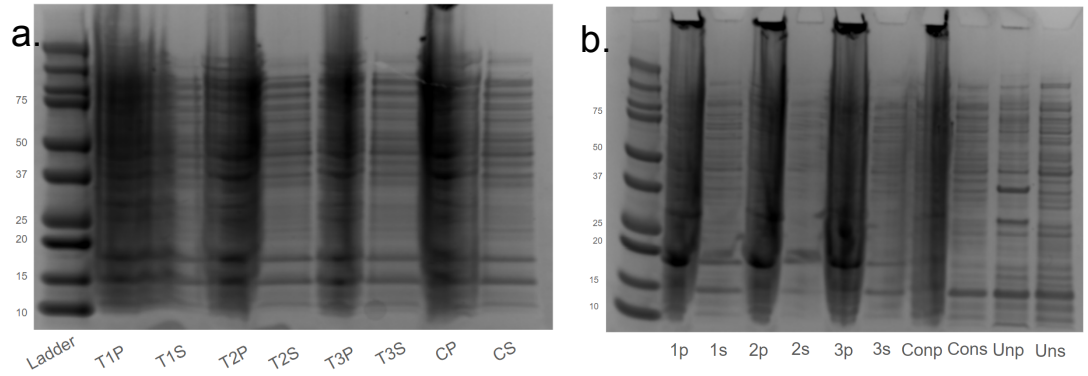


Figure 2: MagLOV is indicated by a strong band at 18kDa a) SDS-PAGE of pRSET cell lysate. T1 was grown with 400mM IPTG and 50 μM Riboflavin, T2 was grown with 400mM IPTG and T3 was grown with 100mM IPTG. C lanes received no IPTG. P visualizes the insoluble fraction of cell lysate and S visualizes the soluble fraction. b) SDS-PAGE of pET Vector transformed cell

lysate. 1 was grown with 400mM IPTG and 50 μM Riboflavin, 2 was grown with 400mM IPTG and 3 was grown with 100mM IPTG. Con lanes received no IPTG. Un lanes received no IPTG and were not incubated at 16°C. P visualizes the insoluble fraction of cell lysate and S visualizes the soluble fraction.

pRSET vector is shown to express MagLOV protein in all lanes, even lanes without IPTG induction such as the CP and CS lanes (Figure 1, a). This gave insight into an issue where cultures would often stop growing after transferring into the secondary growth culture. This issue could have been caused by an effect where T7 promoter plasmids lose their plasmids due to the overactivity—or leakiness—of the T7 promoter (Zhang et.al Protein Expression and Purification 2003). In order to address this, a higher selective pressure was applied to primary cultures so that only E.coli with intact plasmids could survive and propagate into the secondary cultures, which is why pRSET-MagLOV is grown with 5 times the standard working concentration of other ampicillin based expressions. Even with this increased selective pressure though, the cultures would still often die, and yields were quite low for the purposes of crystallization. As such we switched to the pET vector, which our lab had previously optimized to great effect in other protein expressions. Expression in pET shows 18kDa bands indicative of MagLOV in lanes with IPTG—our induction factor—and shows them as absent in lanes without (Figure 1, b), demonstrating a much better regulated T7 promoter in the case of pET vector. Although all purifications after Ni-NTA, and all crystal screen trials were performed using protein expressed from pET vector, we still used pRSET to create some purified protein and plates for MFE check. This was because Calico's studies were performed with pRSET and we wanted to reproduce their results.

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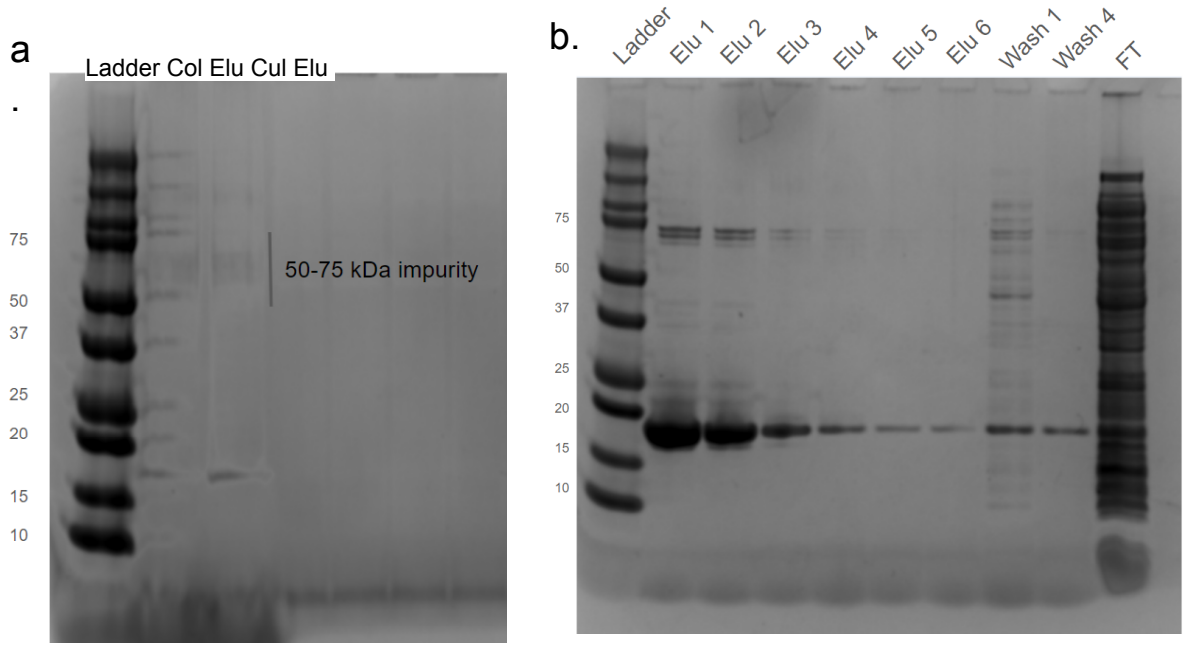


Figure 3: Presence of MagLOV on SDS-PAGE is indicated by an 18kDa band a) SDS PAGE analysis of Ni-NTA on protein from pRSET plasmid expressed by directly placing primary culture or transformed colonies into cultures where they were incubated for 63 hours. Col Elu was elution taken from the culture started from a single colony, Cul Elu was taken from the culture started from a primary culture. The second lane of this gel had an experimental error where the ladder bled into it b) Full SDS PAGE analysis of Ni-NTA on protein from cells transformed with pET Vector. Elu 1-6 are collected elutions of MagLOV. Wash 1 and 4 were taken from the washing step of Ni-NTA, and FT represents all protein in the lysate which did not bind to Ni-NTA beads.

Analysis of the Ni-NTA SDS-PAGE reveals a significant impurity at 75kDa. It was decided this would be purified out using SEC.

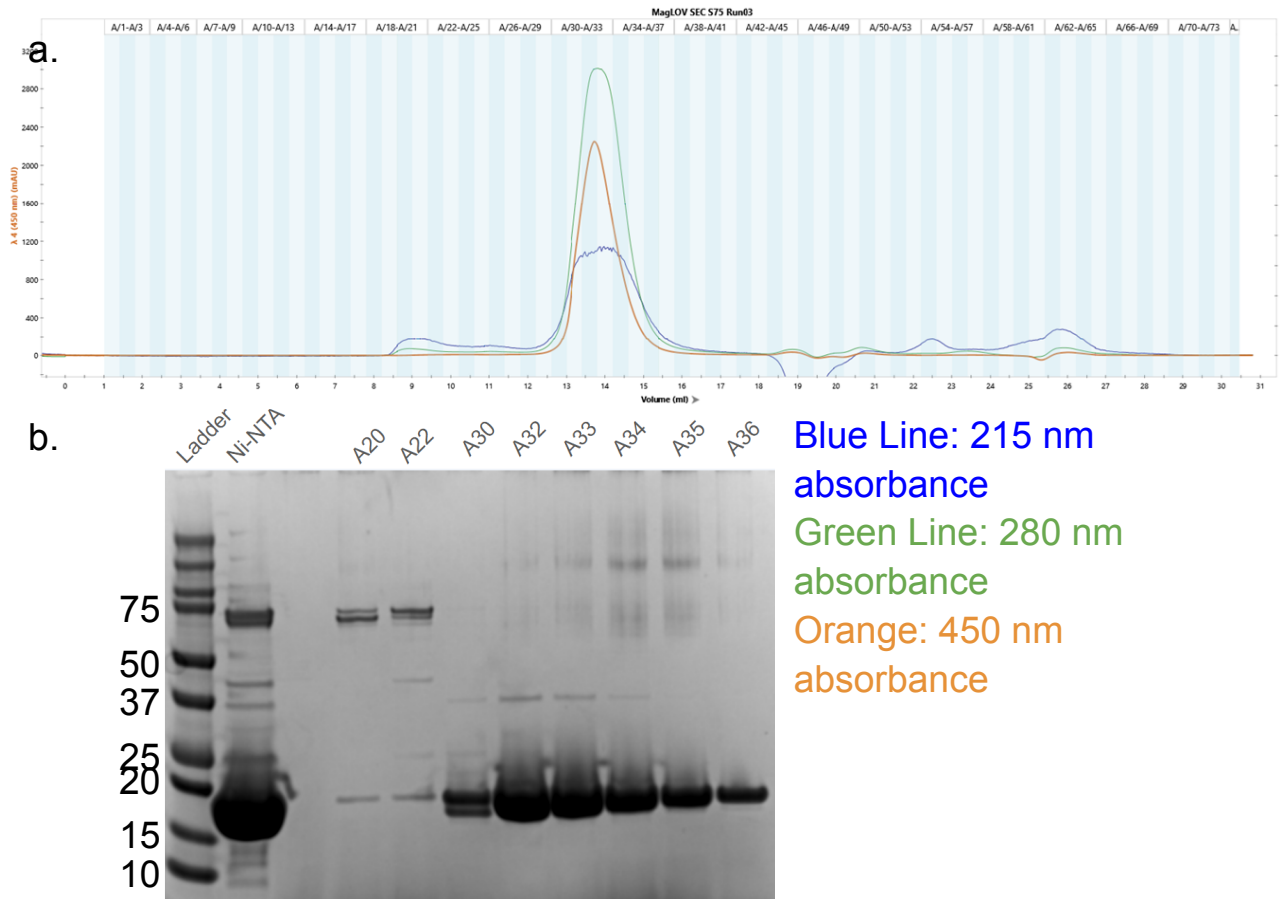


Figure 4: a) SEC Chromatogram on protein expressed from pET Vector and purified by Ni-NTA. Absorbance at 215 and 280 serve as markers for peptide bonds and Trp or Tyr residues respectively. Absorbance at 450 nm indicates FMN and MagLOV, as 450nm is the emission maximum of FMN b) SDS-PAGE loaded with a sample of untreated Ni-NTA protein, wells 20, 22, 30, 32, 33, 34, 35 and 36.

The Chromatogram produced from an SEC of the Ni-NTA elute revealed 4 major peaks. These peaks were located at wells 20-22, 30-36, 54, and 62-65. According to the absorbance, we concluded that wells 20-22 were the 50kDa impurity noted in Ni-NTA SDS PAGES, and that the

30-36 well fractions were MagLOV. The other two peaks were not collected, but likely were results of the high imidazole content of the buffer MagLOV was in after Ni-NTA.

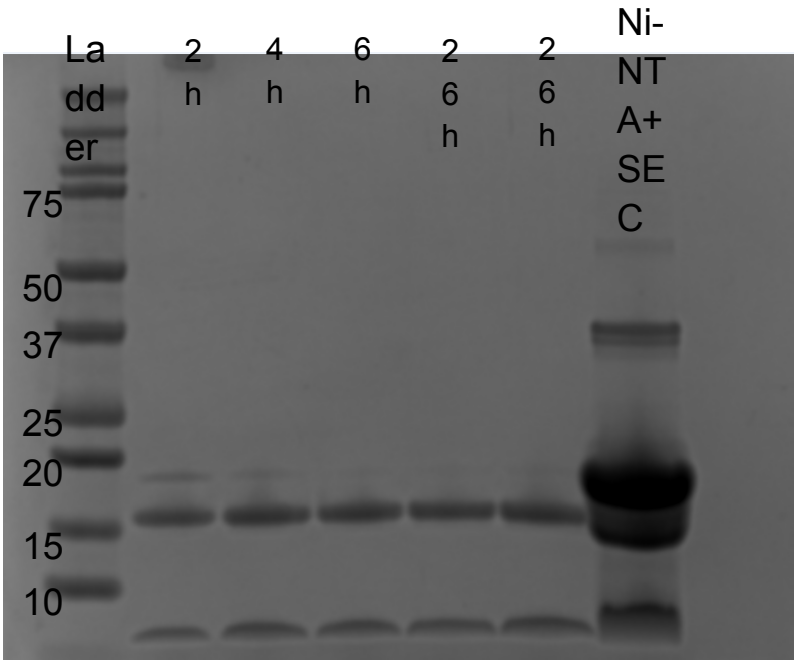


Figure 5: SDS PAGE on the results of a thrombin cleavage performed on pET-expressed protein that had undergone SEC and Ni-NTA. Lanes show protein at different points throughout cleavage. Also loaded is a sample of protein treated with Ni-NTA + SEC without HIS cleavage. Uncleaved MagLOV is present as a 18kDa band, while cleaved MagLOV is present as a 16kDa band.

pET vector expressed protein was collected at various stages in purification. From Ni-NTA+SEC, we have collected around 64 mg of protein. This was split into 50 µL and 10 µL aliquots respectively, flash frozen in dry ice and isopropyl alcohol and stored at -80°C. We've also collected 2mg of NiNTA+SEC purified HIS-Tag cleaved sample This is similarly stored by flash freezing in dry ice and isopropyl alcohol, before being deposited in a -80°C fridge. Finally,

2.5 mg of NiNTA+SEC purified, HIS-Tag cleaved and desalted protein is similarly stored by flash freezing in dry ice and isopropyl alcohol, before being deposited in a -80°C fridge.

B. MFE Experimental Results

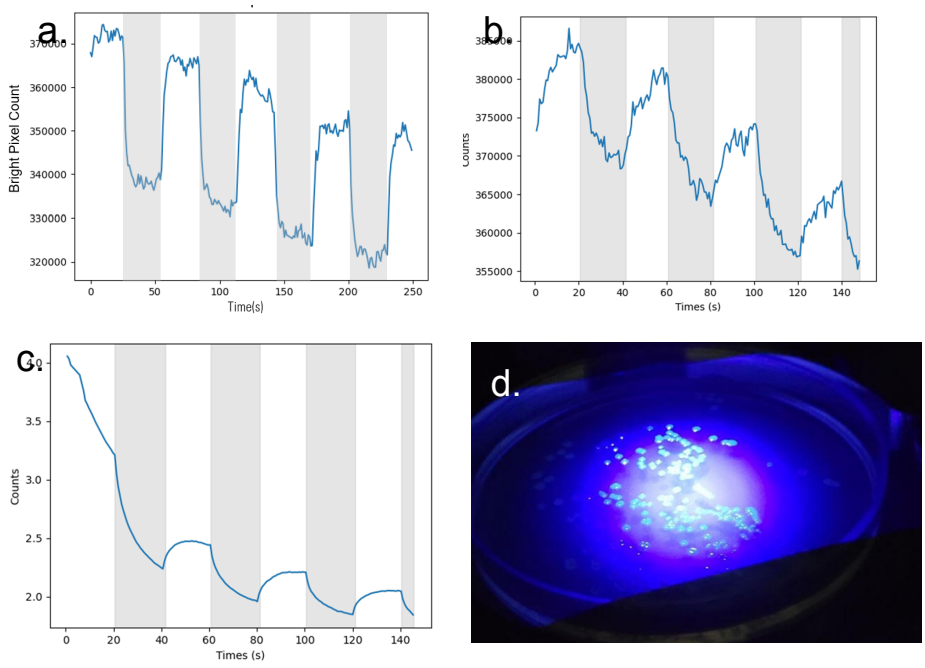


Figure 6: Magnetic field strength is reported as a range due to the fact that the strength of magnetic field is likely highly variable between experiments. a) a picture of pRSET-MagLOV containing colonies on 10μM IPTG 100 μM Amp Plates under blue laser light. b) MFE Check on pRSET-MagLOV containing colonies on 10μM IPTG 100 μM Amp Plates. Magnetic field strength 30-70 mT c) MFE Check on pRSET expressed purified protein (Ni-NTA). Magnetic field

strength 30-70 mT protein concentration: 13μM d) MFE Check on pET vector expressed purified protein (Ni-NTA+SEC+Desalting, No His Tag Cleavage). Magnetic field strength 30-70 mT protein concentration: 91μM

MFE is confirmed in pRSET expressed proteins in both live colonies and in purified protein. MFE is also confirmed in protein expressed from pET vectors.

C. Crystal Screens Show Limited Results

With the crystal conditions based on previous AsLOV2 trials, the 40.4 mg/ml wells all formed precipitate. This suggests that the protein is at least not soluble in those solutions at higher concentrations. The 192 condition crystal screen resulted in some crystal formation. One of the conditions formed a large single crystal (Figure 7) which needs to be confirmed by further reproducibility of the conditions and finally under X-ray diffraction. The 288 well screen tested this same condition without CaCl2 and returned no protein, although some conditions did have phase separation.

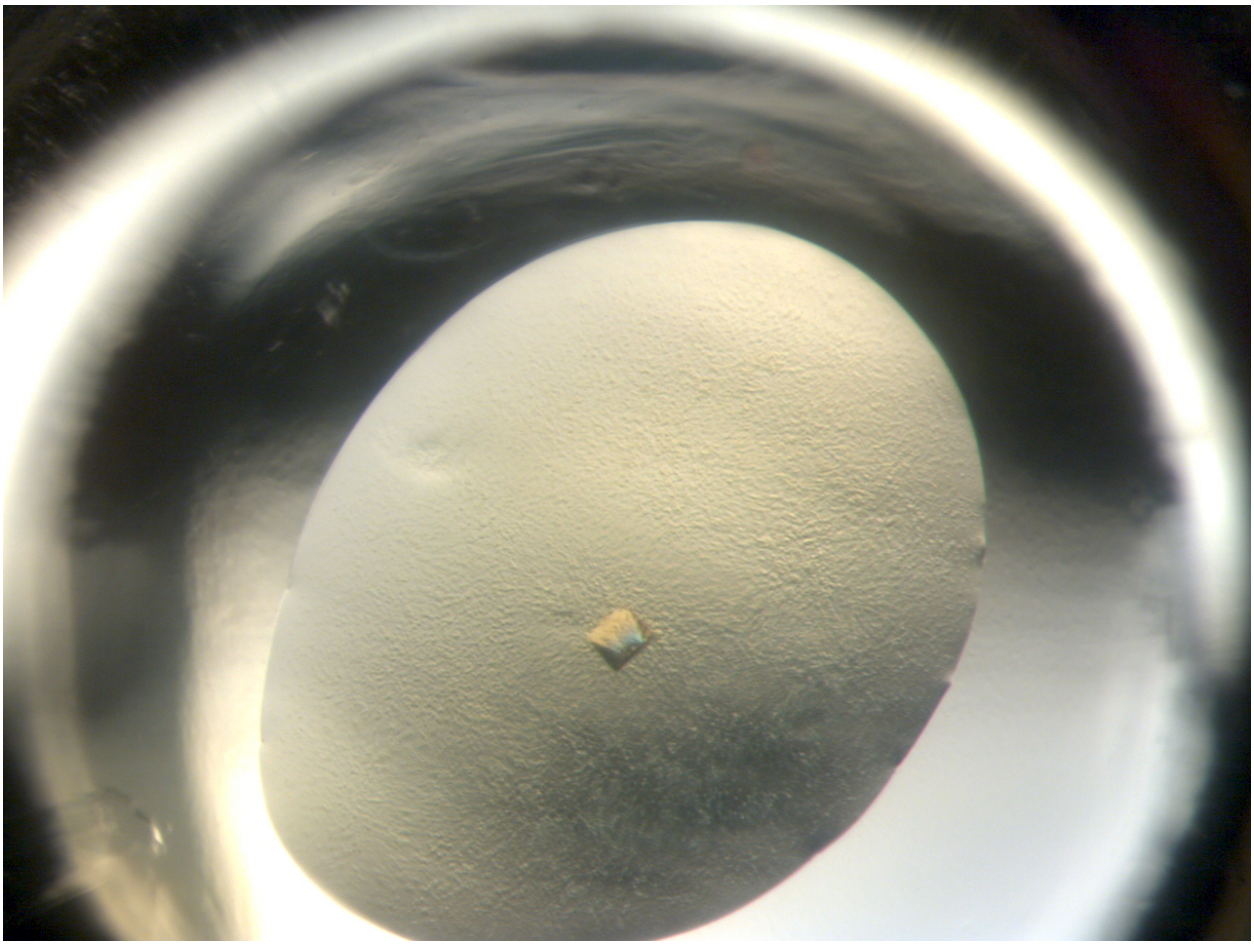


Figure 7: Large single potential MagLOV crystal.

IV. DISCUSSION

After robust MagLOV crystals form, our next goal will be to send MagLOV crystals to an X-ray source to diffract them for a crystal structure of light, dark, MFE affected and MFE unaffected states. Due to MagLOV's status as a LOV domain protein, we hypothesize that these effects could have an associated structural change. If structural change can be identified, the structural data generated from our study will motivate and guide future research to engineer MagLOV in the same way other LOV domain proteins have been engineered. We believe that

the groundwork laid in this study will enable the development of a new type of "magento genetic" tool, acting as the first step in the creation of a novel way to control biology.

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